

PROBABILITY & STATISTICS

Part A — Theory (Short Questions)

An illustrated set of answers with diagrams

“Quality is our Motto.”

Part A — Theory (Short Questions)

Q1. Mean, Median, Mode — Customer Income

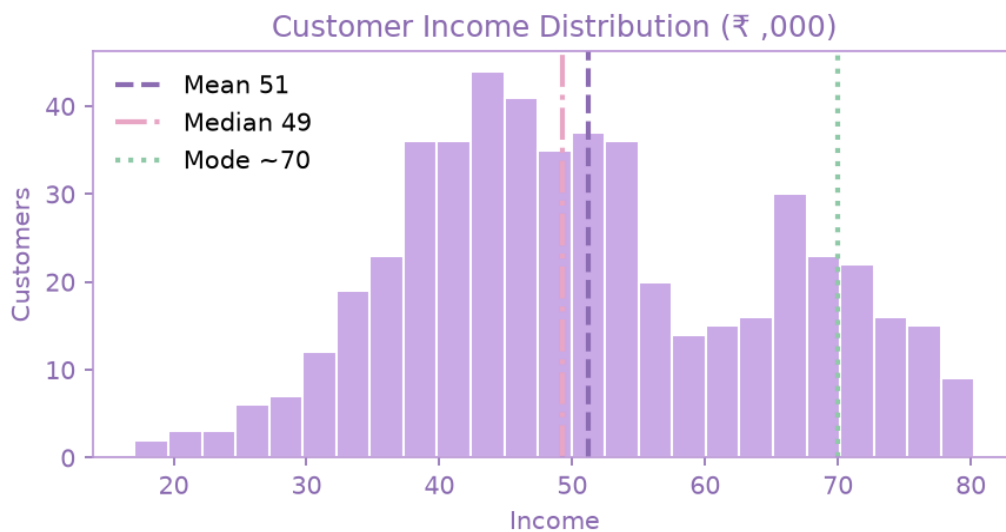
For a bank analysing customer income, the three measures of central tendency describe the "typical" earner from different angles.

Mean: Arithmetic average of all customer incomes; sensitive to a few very-high-income clients.

Median: Middle income when values are sorted; robust to outliers, a better "typical" figure.

Mode: The income value that occurs most often — useful for identifying the dominant customer segment.

$$\text{Mean} = \Sigma x_i / n \quad \text{Median} = \text{middle value} \quad \text{Mode} = \text{most frequent value}$$



Q2. Standard Deviation vs Variance — Loan Amounts

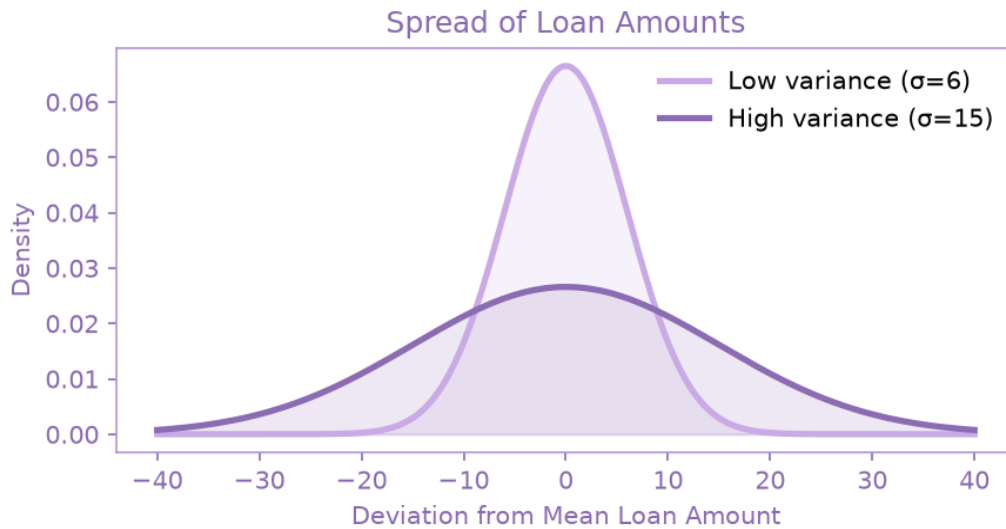
Both measure how much loan amounts vary around the average loan size, but they are expressed in different units.

Variance (σ^2): Average of the squared deviations from the mean loan amount — units are ₹²(squared).

Standard Deviation (σ): Square root of variance — expressed in the same unit (₹) as the loans, so easier to interpret.

$$\sigma^2 = \Sigma (x_i - \mu)^2 / N \quad \sigma = \sqrt{\sigma^2}$$

A small σ means most loans are close to the average; a large σ means the bank issues a very wide range of loan sizes.



Q3. Random Variable

A random variable is a numerical value assigned to the outcome of a random experiment. It can be discrete (countable) or continuous (measured on a scale).

Example from the dataset: Let X = the loan amount (in ₹) of a randomly selected customer. X is a continuous random variable. Another example: Y = number of defaults in a month — a discrete random variable.

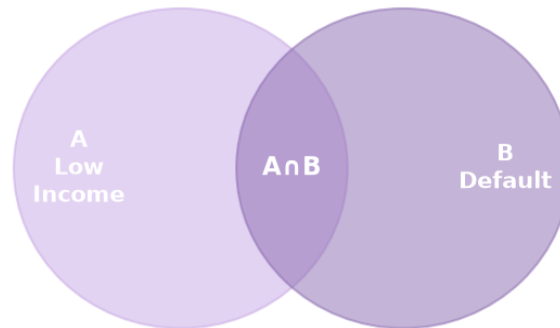
Q4. Conditional Probability — Loan Defaults

Conditional probability is the probability of an event A occurring given that another event B has already occurred.

$$P(A | B) = P(A \cap B) / P(B), \text{ provided } P(B) > 0$$

Example: If A = "customer defaults" and B = "customer has low income", then $P(A|B)$ tells the bank the default rate among low-income customers — a key input for credit-risk scoring.

$$P(\text{Default} \mid \text{Low Income}) = P(A \cap B) / P(A)$$



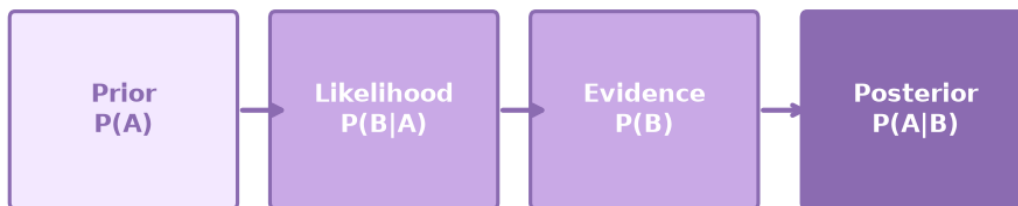
Q5. Bayes Theorem — Application in Banks

Bayes theorem lets us reverse conditional probabilities — updating a prior belief in light of new evidence.

$$P(A \mid B) = [P(B \mid A) \cdot P(A)] / P(B)$$

Bank application: Given a customer defaulted (B), what is the probability they belonged to the "low-income" segment (A)? Banks use this to refine credit-scoring models, fraud detection, and to update risk categories as fresh transaction data arrives.

Bayes Theorem — Updating Beliefs

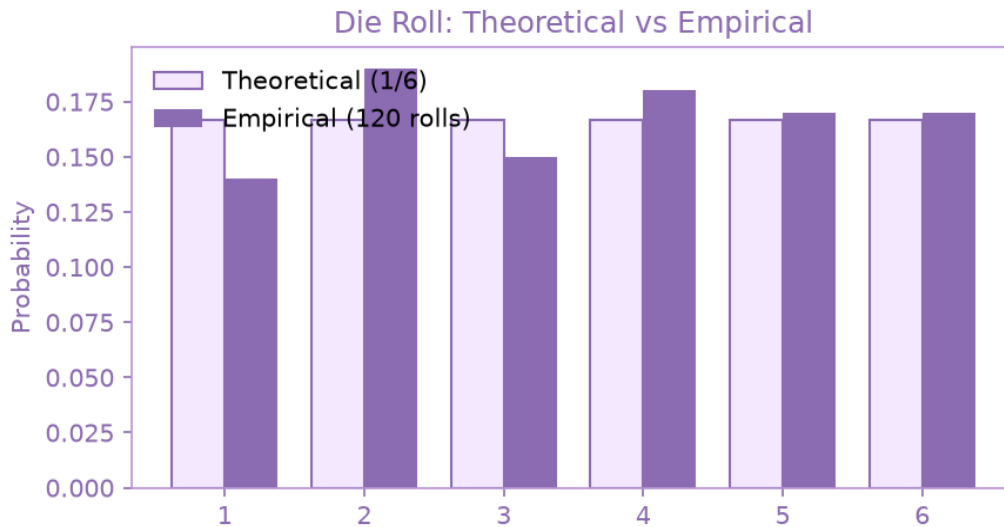


Q6. Empirical vs Theoretical Probability

Theoretical Probability: Based on reasoning about equally likely outcomes. Example: $P(\text{rolling a 4}) = 1/6$ for a fair die.

Empirical Probability: Based on observed data from an experiment. Example: If in 120 rolls the number 4 appeared 22 times, empirical $P(4) = 22/120 \approx 0.183$.

As the number of trials grows, empirical probability tends to converge to the theoretical value (Law of Large Numbers).

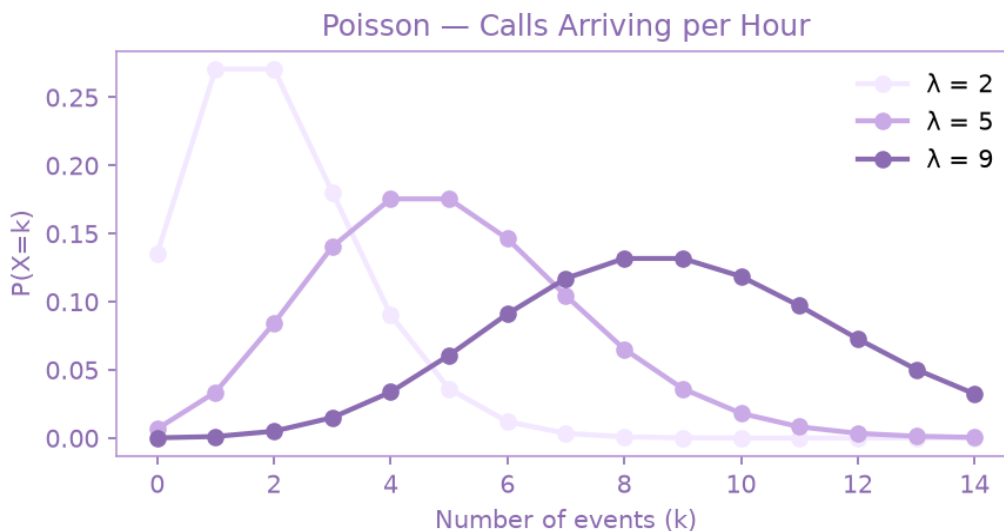


Q7. Poisson Distribution

The Poisson distribution models the number of times an event occurs in a fixed interval of time or space, when events occur independently at a constant average rate λ .

$$P(X=k) = (e^{-\lambda} \cdot \lambda^k) / k!$$

Business example: A bank branch receives on average $\lambda = 5$ customer-support calls per hour. The Poisson distribution gives the probability of receiving exactly 0, 1, 2, ... calls in the next hour — used for staff scheduling.



Q8. Eigenvalues & Eigenvectors in Data Analysis

For a square matrix A , a non-zero vector v is called an eigenvector if $A \cdot v = \lambda \cdot v$, where the scalar λ is the corresponding eigenvalue. Multiplying by A only stretches v (by λ) — it does not change its direction.

$$A \cdot v = \lambda \cdot v$$

In data analysis they are the backbone of Principal Component Analysis (PCA):

- Eigenvectors of the covariance matrix identify the directions of maximum variance in the data (the "principal components").
- Eigenvalues measure how much variance each direction captures.
- Keeping only the top-k eigenvectors reduces dimensionality while preserving most of the information — invaluable for image compression, customer segmentation and noise reduction.

